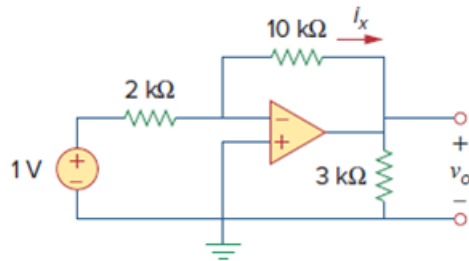


Problem

For the circuit in Fig. 5.40, current i_x is:

- (a) $600\ \mu\text{A}$
- (b) $500\ \mu\text{A}$
- (c) $200\ \mu\text{A}$
- (d) $112\ \mu\text{A}$

Figure 5.40:

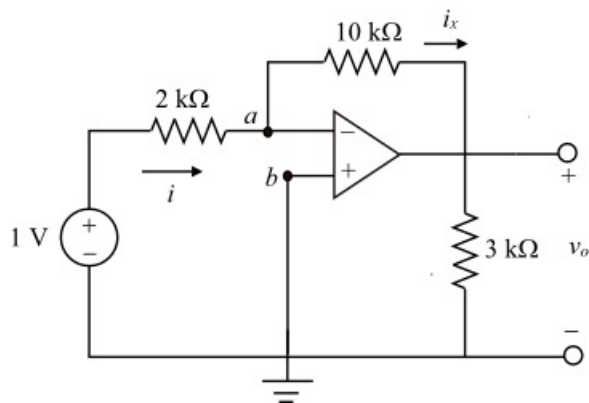


Step-by-step solution

Step 1 of 3

Refer to Figure 5.40 in the text book for the op-amp circuit.

Redraw the circuit by marking nodes a and b .



Step 2 of 3

Apply Kirchhoff's current law at node a .

$$i = i_x$$

Apply Ohm's law to find the current, i .

$$i = \frac{1 - v_a}{2 \text{ k}\Omega}$$

According to virtual ground concept, the voltage at the non-inverting input is equal to the voltage at the inverting input of the op-amp. As $v_b = 0 \text{ V}$, v_a is also equal to 0 V.

Substitute 0 for v_a .

$$\begin{aligned} i &= \frac{1 - 0}{2 \text{ k}\Omega} \\ &= 0.5 \text{ mA} \\ &= 500 \times 10^{-6} \text{ A} \\ &= 500 \mu\text{A} \end{aligned}$$

Hence, the options (a), (c) and (d) are wrong.

Step 3 of 3

The current through feedback resistor is,

$$i_x = 500 \mu\text{A}.$$

Hence, the correct option is **b**.